Cécile **Le Sueur**

Applied Probabilistic Machine Learning Researcher | Technical Storytelling and Science Communication

CecileLeSueur   Scientific Communication Portfolio   0000-0001-7114-3744

France · Open to relocation

About Me

Applied probabilistic machine learning researcher and science communicator with expertise in statistical modelling, technical communication, and scientific storytelling. My work combines mathematical rigour, educational resource development, and interdisciplinary collaboration to make complex ML/AI and statistical concepts clear, engaging, and accessible across scientific audiences, from clinicians to ML engineers. Passionate about conceptual abstraction and education, I thrive when helping people engage with technically intimidating ideas more confidently and thoughtfully without sacrificing precision.

Key Skills

Technical & Research
- Probabilistic machine learning
- Statistical modelling & inference
- ML/AI technical fluency
- Computational workflows & reproducibility

Communication & Education
- Technical storytelling
- Scientific & interdisciplinary communication
- Conceptual abstraction & translation across audiences
- Educational content & resource development
- Visual thinking & data visualisation

Collaboration & Working Style
- Interdisciplinary collaboration
- Audience-centred communication
- Fast learning & intellectual curiosity
- Strategic problem solving
- Project ownership & initiatives

Scientific Communication, Education & Storytelling

Technical Storytelling & Educational Resources
Portfolio ↗

- Translated complex ML/AI and statistical models into conceptually intuitive educational resources bridging biological experimentation, mathematical formulation, and computational implementation.
- Eight years of experience adapting technical communication across interdisciplinary scientific audiences ranging from clinicians and biologists, to computational researchers and ML/AI specialists.
- Created communication resources across multiple formats, including workshops, videos, tutorials, scientific illustrations, interdisciplinary international conference talks & posters, and web-based documentation platforms.

Conferences & Scientific Recognition

- Delivered talks and posters at five international computational biology conferences within two years, including an award-winning presentation; additionally invited to chair a scientific session following recognition by conference organisers.

Selected Scientific Publication
ORCID ↗

- First-author publication on applied probabilistic ML for biological systems in the leading computational biology journal PLOS Computational Biology. *DOI: 10.1371/journal.pcbi.1011632 ↗*

Experience

Editorial workflow optimisation and technical documentation
Apr 2026 – present

EDP Sciences (part-time) — *France*

- Analysed editorial and submission workflows to identify process inefficiencies, introducing digital tools and user documentation to improve usability, onboarding, and day-to-day operational efficiency for editors, authors, and reviewers.
- Developed tutorials, user-facing documentation, and web-based guidance materials to support adoption of reference managers, submission platforms, and publication workflows across audiences with diverse technical backgrounds.

Doctoral researcher

Applied probabilistic Machine Learning for thermal stability proteomics — *Oct 2020 – Jan 2025*

EMBL Heidelberg, European Molecular Biology Laboratory — *Germany*

ETHZ, Swiss Federal Institute of Technology in Zürich — *Switzerland*

- Applied probabilistic modelling and statistical learning methods to complex biological datasets, bridging mathematical theory, ML implementation, and scientific interpretation.
- Developed an open-source analytical pipeline integrating hierarchical Gaussian Process modelling, statistical analysis, and workflow orchestration using Python, R, and Nextflow for reproducible and scalable biological data analysis.
- Designed and developed a web-based platform for scientific knowledge dissemination, translating the theoretical foundations and computational implementation of the analytical workflow into structured, accessible learning resources to support interdisciplinary user onboarding.
- Designed and delivered workshops, scientific presentations, and posters explaining complex statistical and ML concepts to audiences ranging from experimental scientists to computational researchers and ML engineers.
- Developed data visualisations and explanatory materials to communicate complex model behaviour, translate experimental designs into mathematical frameworks, and convey statistical insights clearly across interdisciplinary environments.

Research assistant

Dynamic modelling of tumour evolution and cellular populations — *Oct 2019 – May 2020*

ETHZ, Swiss Federal Institute of Technology in Zürich — *Switzerland*

- Contributed to dynamic modelling studies using stochastic simulation for tumour evolutionary dynamics and compartment-based latent dynamics modelling of treatment-induced hematopoietic cell populations.
- Contributed to model refinement, validation, and scientific publication efforts by producing quantitative analyses and data visualisations to support rigorous interpretation and communication of complex simulation results.

Cross-domain transfer of statistical modelling across biological systems *Feb 2018 – Sep 2019*

CHUV, Lausanne University Hospital & EPFL, Swiss Federal Institute of Technology in Lausanne *Switzerland*

- Led an interdisciplinary research project adapting statistical learning and clustering approaches from fMRI time-series and 3D voxel-based neuroimaging data to multidimensional longitudinal biomarker datasets in precision oncology through abstraction and transfer of modelling principles across distinct biological contexts.
- Bridged computational and clinical research environments by translating mathematical modelling concepts and statistical interpretations across audiences with highly different technical backgrounds.
- Developed analytical approaches for high-dimensional longitudinal and spatiotemporal datasets using unsupervised learning, functional data analysis, similarity learning, and missing-data modelling to improve robustness and interpretability across highly heterogeneous real-world biological data.

Bachelor & Master's researcher (selected projects)

Bayesian inference for proteomics under missing data *Aug 2017 – Feb 2018*

EMBL Heidelberg, European Molecular Biology Laboratory *Germany*

- Developed Bayesian inference and MCMC-based modelling approaches for high-dimensional proteomics datasets by integrating differential expression analysis with probabilistic imputation strategies to improve robustness under uncertainty and missing-data conditions.

Mathematical modelling of human decision-making systems *Feb 2017 – Jul 2017*

EPFL, Swiss Federal Institute of Technology in Lausanne *Switzerland*

- Independently expanded a small experimental social science project into a broader mathematical modelling study combining utility theory, experimental design, and behavioural data analysis, with experimental protocols specifically designed to ensure statistical validity and alignment with downstream modelling requirements.
- Communicated rigorous mathematical modelling concepts within an interdisciplinary social science environment involving audiences with diverse levels of quantitative expertise.

Education

PhD - Probabilistic Machine Learning applied to Proteomics data *Oct 2020 – Jan 2025*

ETHZ, Swiss Federal Institute of Technology in Zürich & EMBL (Joint Degree) *Germany*

MSci - Applied Mathematics and Computational Sciences *Sep 2016 – Jul 2019*

EPFL, Swiss Federal Institute of Technology in Lausanne *Switzerland*

BSc - Life Sciences and Technology *Sep 2013 – Jul 2016*

EPFL, Swiss Federal Institute of Technology in Lausanne *Switzerland*

Awards

- Best **Oral Presentation** Award (CompMS Session, ISMB/ECCB International Conference in Computational Biology, 2023)
- Best Master Project **Poster** Award — Cross-domain transfer of statistical modelling across biological systems (EPFL, 2019)
- **Excellence Fellowship** for academic achievement (EPFL, awarded twice)
- Prize for best average during the first selection year (2014) and second best average across the full bachelor cycle (2016) among 13 scientific programmes at world-leading university (EPFL).

Technical Toolkit

- **Programming & Scientific Computing**: Python · R · Matlab · C++ · R Shiny · HTML
- **Workflow & Development**: Nextflow · Git · Docker · VS Code · Jupyter Notebook
- **Scientific Communication & Visualisation**: Quarto · Inkscape · Affinity Designer
- **Languages**: English (C1) · German (B1) · Spanish (A1) · French (native)

Other Relevant Activities

- **One-year bikepacking journey** across multiple countries (Feb 2025–Mar 2026), combining physical endurance, cultural exploration, adaptability, problem solving, and communication across diverse social and linguistic environments.
- Continuously **exploring new challenges** across physical and mental disciplines, from pushing resilience and endurance through long-distance running to deliberately stepping into intimidating and unfamiliar environments through martial arts and other activities outside my comfort zone.
- Passionate about **dance**, continuously learning new styles and exploring communication and collaboration through solo and partner dancing, as both leader and follower.